PakEmis Sector Technical Note TN02

A Plant-Resolved, Fuel-Constrained Inventory of Greenhouse-Gas and Air-Pollutant Emissions from Electricity Generation in Pakistan

Rehan Ahmad[a,*]

[a] *Pakistan Air Quality Initiative (PAQI), Karachi, Pakistan*

Abstract

This technical note documents the power contribution to PakEmis, a shared greenhouse-gas and air-pollutant inventory for Pakistan covering July 2015–June 2025. The reference uncertainty and detailed spatial year remains FY2023–24. Public-power coal is calculated from station consumption and documented rank-specific net calorific values, while generation, heat rates and the publisher's aggregate energy column provide distinct checks and alternatives. Captive gas self-generation is reported under industrial combustion.

The FY2023–24 power calculation module contributes 45.66 Mt CO_2e on AR5 GWP-100, including captive gas; FY2024–25 contributes 43.86 Mt. Native gas masses and GWP aggregates remain separate. The central air panel admits 30.02 kt of oil/diesel SO_2 in the reference year and 5.707 kt in FY2024–25. Coal and other power pollutants remain separate screening cases because fuel quality and demonstrated control operation are unresolved. Public licence applications establish proposed 92% and 93% flue-gas desulfurisation at Port Qasim and China Power Hub, respectively; these design values support explicit selected-plant scenarios, not measured annual stack emissions. Fuel-specific K-Electric coverage alternatives are published separately. The note provides reproducible activity controls, method boundaries and limitations without treating screening endpoints as mathematical bounds.

Keywords: electricity generation, captive power, energy-basis activity data, coal blend, industrial combustion, control scenarios, AP-42, IPCC 2006 Guidelines, Pakistan, MRV

PakEmis Sector Technical Note TN02. Technical documentation of one sector module of the PakEmis release identified below, deposited with the data as part of the documentation record. The national data-description article (`paper_national/`) is the only document that states the national totals as its own result.

Release provenance. This document describes PakEmis under the period, configuration and source boundaries specified in its methods. The identifiers below link it to the accompanying products and source register.

Release version	`1.0.0rc69`
Commit	`d8d874622f9a`
Reporting period	July 2015–June 2025 (ten fiscal years) (2015-07-01 to 2025-06-30)
Reference	S_TAN (household survey; TAN nitrogen)
FY2023–24 total	444.61 Mt CO_2e (AR5) under S_TAN, the reference; 449.85 Mt under L_TAN
Release manifest	`ac0ffe93ad9fcd88` (SHA-256 of `outputs/release_manifest.json`, first 16)
Source manifest	`c1be6d314cfce1ea` (SHA-256 of `data/manifests/source_manifest.csv`, first 16)
Data-description article	`paper_national/` (ESSD, in preparation), same release

*Corresponding author

Email address: `rehan@pakairquality.com` (Rehan Ahmad)

1. Introduction

Pakistan's thermal fleet combines imported bituminous coal, domestic lignite, gas/RLNG and liquid fuels across CPPA-G/GENCO and K-Electric. NEPRA publishes national fuel controls and plant-level physical fuel, generation and heat-rate records. Their reconciliation provides country-specific activity and boundaries that an aggregate energy table cannot resolve.

This note selects station coal consumption times rank-specific NCV, distinguishes energy from electricity, and tests fuel-specific system coverage. It supports one common GHG and air inventory over July 2015–June 2025. Captive gas is calculated here but reported in industrial combustion. The module's reference-year AR5 total is 45.66 Mt CO_2-equivalent; native gases remain separate.

Reviewed oil/diesel SO_2 joins the central air panel, while coal/gas/control cases remain separately typed. Public design documents improve selected-plant scenarios without supplying measured annual removal. The contribution is traceable station activity, rank/material accounting and explicit boundary alternatives rather than finer nominal grids alone. Shared national methods and products are documented in the national paper (Ahmad, 2026).

2. Related Work

2.1. Fuel-based and generation-based power inventories

Power-sector emissions can be built from fuel consumed or from electricity generated with an assumed heat rate. The IPCC 2006 Guidelines describe the fuel route as Tier 1 and treat generation-based estimates as a fallback where fuel data are unavailable (IPCC, 2006), because the heat rate is an efficiency assumption that varies by plant, by load and by year, and because it is precisely the quantity a compiler is least able to verify.

Pakistan's regulator publishes fuel directly, so the fallback is unnecessary here. This is worth stating because a workbook consolidating Pakistani plant-level data—which supplied the captive register used in this paper—computes its CO_2 as kilograms per megawatt-hour derived from IPCC factors through an assumed plant-efficiency range. That is the generation route, and its output is not used here for that reason, while its plant register is.

2.2. Coal quality and why the unit matters

The fleet burns imported bituminous coal, at Sahiwal, Port Qasim and Hub, and domestic lignite, at Lakhra in the early panel years and from the Thar field thereafter. Published proximate analyses of Thar coal report moisture near 46 %, ash around 6.4 % and calorific values near 3030 kcal/kg—roughly half that of imported steam coal.

That difference is the whole issue. Emission factors expressed per tonne carry it in full; factors expressed per terajoule do not, because the calorific value is exactly what the differing energy content describes. The methodological literature on coal inventories treats blend uncertainty as a major term, and it is—on a mass basis. Section 4 shows it is a 6.8 % term on an energy basis, which changes what a compiler should worry about.

2.3. Captive gas self-generation and the industrial-combustion gap

Industrial self-generation sits awkwardly in inventory practice. IPCC assigns it to 1A2 with manufacturing-industry factors, but the data that would support it are usually collected—if at all—by an electricity regulator whose remit stops at plants selling into the grid.

For Pakistan the consequence is visible in the international comparisons: EDGAR (Crippa et al., 2025) reports Pakistani industrial combustion near 44 Mt while most national compilations report nothing, and the difference between a regulator's scope and an inventory's scope is a substantial part of that. The activity data that do exist come from a different regulator entirely—the oil and gas authority, which reports gas delivered to captive power because it regulates the pipeline rather than the plant.

2.4. Uncontrolled factors and control-device inventories

US EPA's AP-42 compilation (US Environmental Protection Agency, 1995) publishes emission factors for coal, oil and gas combustion predominantly as *uncontrolled* values, with separate treatment of control efficiencies, because what leaves a furnace and what leaves a stack differ by up to two orders of magnitude for particulate and by most of the mass for sulphur.

Applying an uncontrolled factor to a controlled plant requires an explicit operating assumption. The separate cases retain typical uncontrolled and transferred controlled-operation factors, without claiming that actual releases must lie between them. The central admission register and public design evidence use distinct evidence criteria; neither coincidence of two scenarios nor an installed-device claim proves demonstrated annual removal.

3. Scope and activity data

The common inventory covers ten fiscal years, July 2015–June 2025. NEPRA plant tables provide physical fuel consumption, generation and reported heat rates for named stations. The station ledger distinguishes CPPA-G/GENCO and K-Electric, fuel, coal origin, units and publication table. The latest core year uses the FY2024–25 tables in the State of Industry Report 2025. Later generation-performance editions are independent controls where their periods and system boundaries match.

Coal tonnes consumed are distinct from receipts, inventory change, generation and heat-rate-derived mass. The reference selects physical station consumption and rank-specific NCVs. The published national coal-energy series remains a separate control and structural alternative rather than an additional fuel input. Pre-ledger years retain their documented reconstruction and weaker evidence. Facility commissioning and closure delimit source allocation.

OGRA's annual sector deliveries supply captive gas self-generation: FY2024–25 is 266 MMCFD, the sum of the published SNGPL and SSGC categories. This is gas delivered to industrial self-generation, not all industrial combustion or a complete census of captive fuels. A held administrative plant register supplies geographic support; its fuel sums do not establish annual operation of every listed plant.

Monthly fuel and generation tables provide source-appropriate timing controls. Extraction quality and geographic completeness are tested on the exported records. A twelve-month vector is not national observed activity merely because twelve rows exist. Station-level air admission uses physical oil/diesel mass where available and explicitly identified generation/heat-rate reconstruction for the early years.

4. Methods

For station i, fuel rank r and gas g, coal emissions are

$$E_{g,t} = \sum_{i,r} M_{i,r,t}\, NCV_r\, EF_{g,r}/10^6, \tag{1}$$

where M is fuel consumed in tonnes, NCV is TJ per tonne and the factor is kg per TJ; emissions are Gg. Imported bituminous coal uses 25.8 TJ/Gg and lignite 11.9 TJ/Gg. Their carbon

factors are 94 600 and 101 000 kg CO_2/TJ, respectively. Rank energy is calculated before factors are applied. A generation-weighted rank share does not weight fuel carbon.

Rank	NCV (TJ/Gg)	CO_2 factor (kg/TJ)
Other bituminous	25.8	94 600
Lignite	11.9	101 000

Table 1: Rank-specific reference defaults. Physical station mass is retained separately from heat and electricity.

Other grid fuels retain their documented national fuel-energy controls and units. Generation and heat rate test fuel plausibility and provide separately labelled alternatives; they do not replace available physical coal consumption. The station-rank reference, fuel-property alternative, legacy published coal conversion and K-Electric coverage alternative are distinct structural cases without assigned probabilities.

Captive gas volume is converted from MMCFD to annual standard volume, then to net fuel energy using the documented gross heating value and gross-to-net ratio. Captive emissions are assigned to IPCC 1.A.2. Main-activity generation is 1.A.1.a, so the same captive fuel is not added again to public power. Combustion factors already incorporate the IPCC oxidation convention; no additional oxidation discount is applied.

K-Electric coverage is tested fuel by fuel against plant and national controls. The reference CPPA/GENCO boundary and the additional-KE case are published separately where the national table's membership remains ambiguous. Annual native CO_2, CH_4 and N_2O masses are retained, and AR5/AR6 aggregates are computed afterwards.

5. Emission factors and conditional uncertainty

Public generation uses the IPCC stationary-combustion energy-industry factors; captive industrial self-generation uses the manufacturing category. The natural-gas factors are 56 100 kg CO_2, 1 kg CH_4 and 0.1 kg N_2O per TJ in both categories. Coal carbon factors differ by rank. Fuel consumption, rank NCV and factor uncertainties are represented as separate shared parameters, so the same physical fuel draw reaches native gases and applicable air calculations.

The reference coal heat share is the sum of station mass times rank NCV. The historical generation-share series remains a diagnostic and is not an observed fuel-energy share. Rank, moisture/calorific basis and unresolved origin are recorded in the coal account. Alternative fuel-property and coverage cases are not percentiles around the reference.

The tonne-of-oil-equivalent conversion is 0.041868 TJ/t. Its compatibility with the source convention remains relevant for the aggregate non-coal controls. The station-mass coal reference does not inherit the publisher's disputed aggregate coal conversion. For gas, gross heating value and gross-to-net conversion remain explicit conditional assumptions.

Air factors preserve AP-42 source equations and particulate definitions. Sulfur-scaled factors use fuel sulfur once; a controlled factor is not discounted a second time. Unmeasured sulfur specifications and control-operation alternatives are structural cases. The central conditional intervals exclude missing source categories, alternative control configurations and spatial/temporal allocation uncertainty.

6. Source controls and evaluation

6.1. Fuel and generation controls

The final station ledger is checked for unique plant–fuel–year records and compatible units. Plant generation sums are compared with separately read system totals. Fuel mass, NCV-derived heat and electricity imply an efficiency diagnostic, with rank-specific heating values retained. An incompatible publisher coal-energy control is exposed rather than reconciled by changing station consumption. Physical station fuel and national aggregate energy are separate observations with different uncertainty and boundary support.

6.2. National and external comparisons

National reporting, EDGAR and Climate TRACE are compared only at matched periods, source scope and gas conventions. Shared energy-balance inputs reduce independence. Agreement with a filing compiled from the same balance cannot validate the disputed calorific conversion, and a difference from EDGAR does not measure missing fuel.

6.3. K-Electric membership

CPPA/GENCO, K-Electric and national fuel totals are compared for gas, oil and coal separately. Membership inferred for one fuel is not imposed on the others. The FY2024–25 K-Electric gas/RLNG volumes are converted from the reported gross-heat basis to the same net calorific basis used for station combustion before calculating either the coverage ratio or the additional-KE alternative. Oil and coal retain their own physical fuel bases. The supported reference boundary and additional-KE alternative remain separate, preventing duplicated electricity or fuel.

6.4. Satellite evidence

Held satellite products may describe regional and seasonal patterns near facilities. They are column retrievals affected by transport, chemistry, cloud sampling and retrieval sensitivity. A concentration–emission correlation alone is not an emission validation, and the reported fuel/control scenarios are not tuned to satellite columns. A plume inversion or atmospheric-model comparison is outside the present inventory's completion scope.

7. Results

The power calculation module contributes 45.66 Mt CO_2e in FY2023–24 and 43.86 Mt in FY2024–25, on AR5 GWP-100. These module totals include captive gas self-generation, which is reported in the industrial IPCC category. They must therefore not be added to a separate industrial total that already includes captive gas.

The common annual series covers July 2015–June 2025. Native gas, source category and configuration remain queryable independently. Historical differences from the preserved release are recorded with the coal-booking revisions; source growth and accounting changes are not conflated.

7.1. Structural configurations

Station rank-default consumption is the reference. The plant fuel-property case tests calorific assumptions, the legacy energy column tests the superseded publisher conversion, and the additional-KE case tests a coverage boundary. These alternatives answer different questions and have no assigned probabilities. Conditional intervals refer to the selected configuration, not the range across these cases.

Detailed provincial, district and cell records carry placed plus unallocated mass. Plant coordinates and commissioning dates determine the supported point allocation; listed facilities do not imply observed grid-cell activity. Reference-year maps are proxies where own-year operation/location is unavailable.

8. Air-pollutant estimates and separate control scenarios

The central national air panel admits oil/diesel SO_2 from a reviewed plant–fuel–year register. It contributes 30.02 kt in FY2023–24 and 5.707 kt in FY2024–25. Fuel mass, equipment-specific factor applicability, sulfur basis, assumed removal and reporting boundary are recorded separately. The FY2024–25 admitted RFO quantity is 95.11 kt against a compatible 102.6 kt national control. K-Electric remains a separate coverage case.

Coal, gas and bagasse pollutant scenarios do not enter those central totals. AP-42 bituminous/lignite SO_2 factors are $38S/30S$ lb per short ton, with sulfur in weight percent. Filterable particulate factors are $10A/6.5A$; size fractions use the applicable source tables. Rank fuel energy comes from the same physical coal account as GHG. Thar sulfur spans 0.26–2.90%; this range remains an explicit imported-coal proxy where no consignment analysis is held. It is not a measured imported composition or a probability distribution.

Public generation licence applications provide plant-specific *design* evidence. Port Qasim's application (PDF page 69) specifies limestone–gypsum wet FGD at no less than 92%, five-field ESP bulk-ash removal of 99.7%, low-NOx burners and planned monitoring. CIHC/China Power Hub's application (PDF page 204) specifies seawater FGD at no less than 93%. These documents do not establish actual annual removal, bypass or availability. Bulk ESP efficiency is not transferred to $PM_{2.5}$.

Species (Gg)	FY2023–24	FY2024–25
$PM_{2.5}$	NE	NE
BC	NE	NE
NO_x	NE	NE
SO_2	30.02	5.707
CO	NE	NE
NMVOC	NE	NE
NH_3	NE	NE

Table 2: Central power air admission. NE means no central estimate; separate screening cases do not fill these cells.

The separate selected-plant SO_2 cases apply the design FGD efficiency once to station coal consumption and an explicit sulfur proxy. The uncontrolled and design-operation results are conditional scenarios, not central releases, mathematical maxima/minima or 95% confidence limits. Generic controlled coal/gas scenarios remain technology transfers where plant-specific operation is unresolved. Captive gas is counted once in industrial air emissions.

Monthly timing for admitted stations is labelled a parent-fuel-fleet proxy when only aggregate monthly fuel is available. The admitted stations use their own spatial frame. A national fleet profile is not observed timing or geography for a subset.

9. Interpretation

The contribution is the reconciliation of physical station fuel, coal rank, generation and reporting boundaries. A single national energy column hides the rank and calorific assumptions that matter to coal tonnes. Retaining these quantities separately permits source-specific GHG and air methods and shows which differences arise from activity, factors, controls or coverage.

The selected coal reference is defensible without proving that its default NCVs are measured for every consignment. Likewise, public plant design evidence improves control scenarios without establishing actual annual stack removal. More detailed grid cells do not resolve those source uncertainties. The central air estimate and the separate coal/gas scenarios should be used for their stated purposes and never combined silently.

Captive self-generation belongs to industrial combustion. Publishing it in a power calculation module is an implementation convenience, while the category account prevents double counting. The fuel-specific K-Electric comparison illustrates why a single blanket coverage assumption would obscure important source boundaries.

10. Limitations

Station fuel consumption is stronger evidence than a generation-derived estimate, but coal origin, moisture and consignment NCV are not measured comprehensively. Default rank NCVs and carbon factors remain conditional parameters. National receipts, stock changes and consumption are distinct

quantities; unresolved differences are not automatically assigned to a consumer.

The source convention for aggregate non-coal energy and gas gross-to-net conversion is imperfectly documented. K-Electric membership remains fuel-specific where compatible controls cannot select a unique boundary. The selected reference and a separate coverage alternative expose the consequence rather than adding both.

Captive gas is one supported fuel/use. Other captive fuels and the industrial engine/turbine partition remain incompletely observed. The national industrial account supplies other admitted fuel uses; the difference from a global inventory is not a floor on missing combustion.

Central power air coverage is deliberately narrow: oil/diesel SO_2 only. Coal sulfur/ash analyses and demonstrated annual control operation are missing. Licence-application FGD and ESP values describe design, not verified performance. Screening endpoints are not bounds and are not included in conditional central percentiles. Particle size and filterable/condensable definitions limit cross-source comparisons.

Early fuel profiles and newer-year geography can be reconstructed or carried proxies. Facility status, source-period evidence and unallocated mass are preserved. Monthly grids represent allocated source mass; they do not supply vertical injection, detailed chemical speciation or a fully evaluated atmospheric-model input.

11. Conclusions

The maintained power inventory uses station coal consumption and rank-specific NCVs, preserves aggregate controls as explicit alternatives, and distinguishes public power from captive industrial fuel. The supported annual core reaches June 2025. Native GHG masses, GWP aggregates and air-pollutant masses retain their distinct conventions.

Reviewed oil/diesel SO_2 enters the central air panel. Coal and other power pollutant/control cases remain separate because plant design does not demonstrate operation and fuel quality is incompletely observed. The release provides a source-specific basis for reviewing those limitations without hiding them in an uncontrolled endpoint or an undifferentiated national total.

Code and data availability

The calculation engine, the activity data, the captive plant register and every output quoted here are version-controlled in the inventory repository (https://github.com/PakAirQuality/pakistan-emissions-sectors) and reproduce from a single command. The sector is regenerated with `python3 -m pkinv.sectors.power.run_power`; Appendix A gives the full sequence. Every number printed in this paper is read from a generated CSV rather than transcribed, and a test asserts that the manuscript and the engine agree.

References

Ahmad, R., 2026. PakEmis 1.0.0rc69: greenhouse gas and air pollutant emissions in pakistan, july 2015–june 2025. National data-description manuscript with technical report and sector notes; local review package prepared. External repository review deposit and persistent identifier pending; no journal submission asserted.

Crippa, M., Guizzardi, D., et al., 2025. GHG emissions of all world countries — 2025 report (EDGAR GHG booklet). European Commission, Joint Research Centre. Sheet GHG_by_sector_and_country; annual series 1970–2024; CC BY 4.0. Snapshot archived at `sources/EDGAR/EDGAR_2025_GHG_booklet_2025.xlsx`, sha256 `8bd6b8a4...` pinned per benchmark row in `data/benchmarks.csv`. https://edgar.jrc.ec.europa.eu/.

IPCC, 2006. 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Volume 2: Energy, Chapter 2: Stationary Combustion. Technical Report. Institute for Global Environmental Strategies, Hayama, Japan. URL: https://www.ipcc-nggip.iges.or.jp/public/2006gl/vol2.html.

US Environmental Protection Agency, 1995. AP-42: Compilation of Air Emissions Factors, Fifth Edition, Volume I, Chapter 1: External Combustion Sources. Technical Report. Office of Air Quality Planning and Standards, Research Triangle Park, NC. URL: https://www.epa.gov/air-emissions-factors-and-quantification/ap-42-compilation-air-emissions-factors. sections 1.1 bituminous and subbituminous coal, 1.3 fuel oil and 1.4 natural gas. Individual sections carry their own revision dates and are revised independently of the edition; the sections used here should be version-locked before final submission.

Appendix A. Reproduction

The sector runs on Python 3.11 with no compiled dependencies:

```
pip install -e .
python3 -m pkinv.sectors.power.run_power
python3 -m pkinv.sectors.power.air_pollutants
```

which writes `power_totals_fy2023_24.csv`, `power_by_stream_fy2023_24.csv`, `power_fuel_fy2023_24.csv`, `power_provincial_fy2023_24.csv`, `power_timeseries_fy2023_24.csv` and `power_air_pollutants_fy2023_24.csv` to `outputs/`.

The run also writes the evidence tables that Section 6 and Section 7 draw on:

- `power_energy_closure_fy2023_24.csv`: the energy closure, each item with the evidence it rests on.
- `power_fuel_fy2023_24.csv`: fuel by stream, and `power_coal_blend_fy2023_24.csv` the blend within coal.

- `power_gas_technology_fy2023_24.csv`: the gas fleet by technology bucket, with each bucket's energy, share and plant count.
- `power_bagasse_fy2023_24.csv`: the bagasse sub-ledger.
- `power_heat_rate_fy2023_24.csv`: the heat-rate series by fiscal year and fuel—generation fuels, TOE, net GWh, implied heat rate and efficiency—with a verdict on each row.
- `power_heat_rate_stability_fy2023_24.csv`: whether each fuel's efficiency is stable across the material years, as a low–high spread.
- `power_hr_circularity_fy2023_24.csv`: the circularity check: the ledger's own Btu/kWh against each plant's reported gross and net heat rates, so a heat rate derived from the ledger cannot be offered as evidence for it.
- `power_boundary_audit_fy2023_24.csv`: the boundary audit—gas and furnace oil with and without Karachi Electric, and K-E's own energy—and `power_ke_leg_fy2023_24.csv` that utility's leg by year with its implied net efficiency.
- `power_lmdi_fy2023_24.csv`: the decomposition of each year-on-year change into activity, mix and factor terms, with the coal, furnace-oil and gas mix contributions separated.
- `power_point_sources_fy2023_24.csv`: the point-source register: plant, fuel and energy with the town and province, and how each was located.
- `power_climatetrace_plants_fy2023_24.csv`: the Climate TRACE plant crosswalk, that inventory's asset against this one's ledger plant with the ratio and any flag raised.
- `power_tropomi_crosscheck.csv`: the satellite cross-check: observed NO_2 and SO_2 enhancement against the ledger's net generation, with the days and months behind each.
- `power_monthly_fy2023_24.csv`: the twelve-month layer—category, subitem and gas by month with each month's share, the emissions it implies in both common-GWP vintages, and the temporal method and basis behind the shape.
- `power_uncertainty_fy2023_24.csv`: the sector interval by category—the AR5 total with its Approach-1 percentage computed both ways, grouped and independent, so the correlation assumption is visible rather than chosen silently.

Figures are regenerated with `python3 -m pkinv.report.figures` and copied into this manuscript by `python3 scripts/sync_figures.py`; a test fails if any shipped figure differs from the output it plots. The full suite runs with `python3 -m unittest discover tests`.

Appendix B. Sources

Grid coal is NEPRA's plant-wise station consumption from the State of Industry Reports, converted with rank-specific net calorific values; the national fuel-energy table (Table 3) supplies gas, RLNG, furnace oil and diesel and is kept as a comparator for coal. Captive gas is the OGRA State of the Regulated Petroleum Industry annual reports for FY2015–16 to FY2024–25. The captive plant register combines the SNGPL captive fleet, the SSGC geocoded captive list and the NEPRA generation-CPP licence register, deduplicated; plant-wise generation for the provincial proxy comes from the same consolidated workbook. All are archived under `sources/Power Plants/` and described in `docs/sources.md`.

Emission factors are IPCC 2006 Guidelines Volume 2 Chapter 2 defaults (IPCC, 2006): Table 2.2 for grid generation, Table 2.3 for captive. Air-pollutant factors are US EPA AP-42 Sections 1.1 (bituminous coal), 1.3 (fuel oil), 1.4 (natural gas), 1.7 (lignite, whose share of coal energy follows station mass times rank NCV), 1.8 (bagasse) and 3.1 (stationary gas turbines), transcribed with their table numbers and quality ratings; the fetched section PDFs are hashed in `sources/AP42/`. External comparison is EDGAR (Crippa et al., 2025).

Appendix C. Supplementary Note: replacement of the provincial allocation proxy

Earlier revisions allocated grid emissions to provinces by installed capacity from a global power-plant database. That proxy predated most of the coal build-out (21.5 GW listed against roughly 45 installed) and assigned provinces by plant coordinates, which substantially overallocated to Khyber Pakhtunkhwa—a province whose electricity is almost entirely hydro, which burns nothing and belongs nowhere in a combustion sector.

The first replacement used contemporaneous plant-wise thermal generation: Punjab 49.67 %, Sindh 34.22 %, Balochistan 16.09 %, Khyber Pakhtunkhwa 0.02 % on FY2022 output, against the capacity proxy's 32.95, 43.61, 9.20 and 14.24 %.

That share was still one geography for every fuel. The current allocation is per fuel, from the FY2023–24 plant fuel ledger joined to the report's own plant-location table (Table 5), energy-weighted with each row's evidence class carried: coal Punjab 9.7 / Sindh 89.4 / Balochistan 0.8 %; gas and RLNG 59.8 / 22.8 / 17.4 %; furnace oil and diesel wholly Punjab in the year's data (Hub reports zero fuel). K-Electric plants are excluded from the shares; whether the national fuel rows already contain their fuel is not established (Section 6.3), and the coverage alternative prices it beside the reference. Bagasse rides its own all-Punjab row. The aggregate FY2022 share remains in the parameter file as the documented fallback, and both replaced proxies' provenance and caveats are preserved in its history. Each committed per-fuel share is recomputed from the ledger by a test and fails on drift.